

Step 5 : Create a set of selection criteria in order to compare the various alternatives

Scale 1-5 (5 = best)

Assessment criteria	Concept (1)	Concept (2)	Concept (3)
Performance (effectiveness)	4-5	4-5	2-3
Price attractiveness	1-2	1-2	4-5 (highest)
Safety	1-2	5	5

Step 6 : Use the selection criteria to assess/rank these concepts
Scale : 1-5 (5 = best or most attractive)

Rank : 1 : Concept (2)
2 : Concept (3) ?
3. : Concept (1) ?

⇒ Use Concept (2) for further development :
build prototypes,

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Agenda:

0: Review the new product design process
& complete the Nasal Opener example
(done; see above)

1: Another example of product conceptual design

2. Product strategy

3. Vision, Mission, and Business Goals for the start-up.

1. Another example of Product Conceptual Design

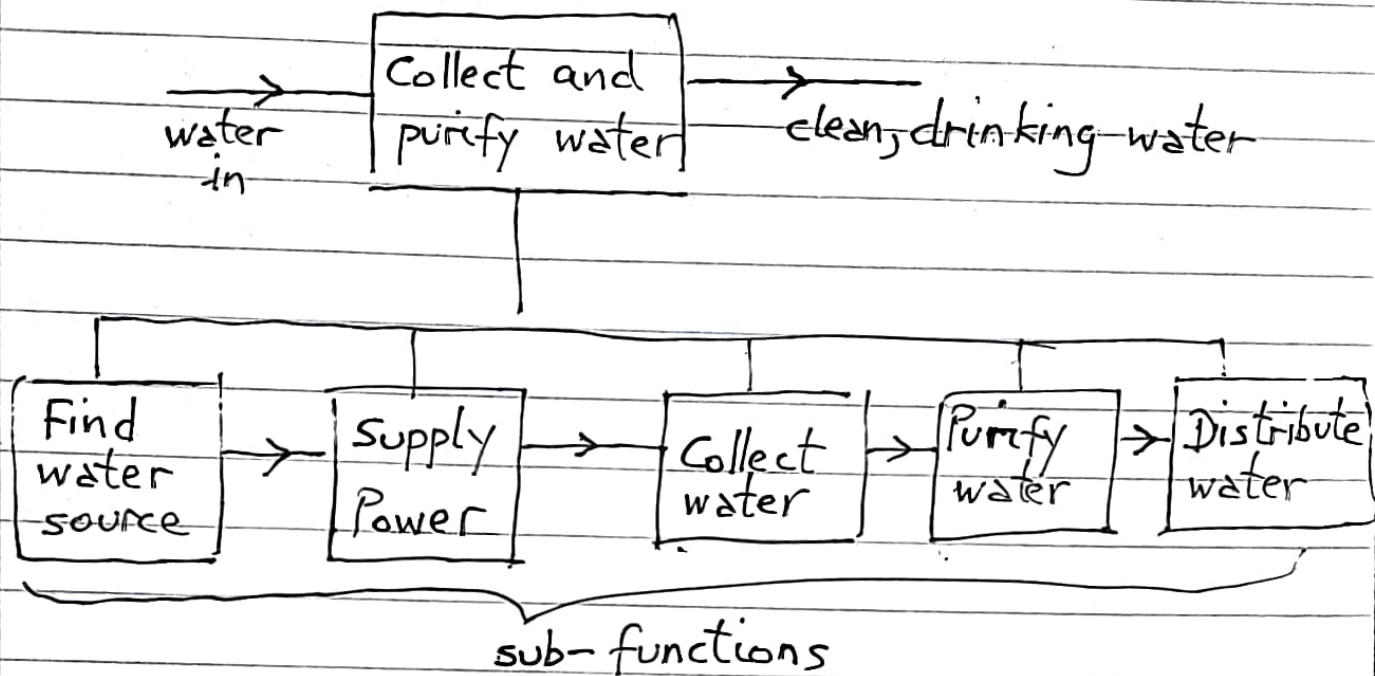
Step 0 : Societal or Customer Need: Clean drinking water for everyone (the planet)

Step 1 : Primary function
Collect & purify water

Step 2 : Create a function structure (FS) for the product

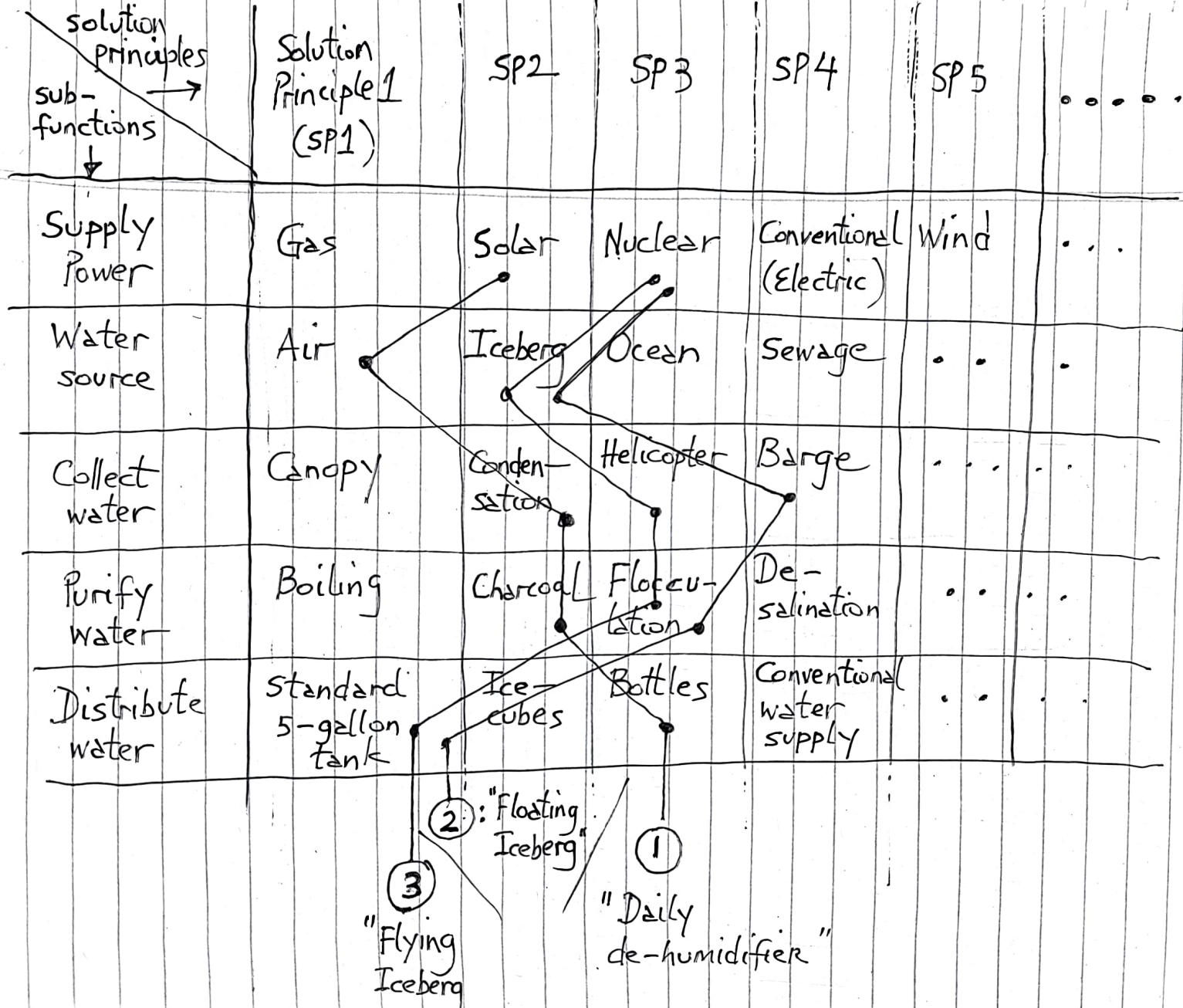
- Look at existing related products (distillation plants, water collection), & dissect these products using FAST
- Use these FAST diagrams to identify key sub-functions in order to create a function structure for the new product.

FS for the new product



Step 3 : Morphological Matrix (MM)

- Write down the sub-functions in column 1.
- For each sub-function (or row of the matrix), generate alternatives or solution principles (SPs) to realize that sub-function



Step 4 : Generate 3-5 alternative design concepts.

See previous page (above) : "Zig-Zag" through the MM to generate these concepts.

Describe how each concept works with simple sketch, & create a FAST diagram for each realized concept.

(sub-function = WHY? ; solution principle = HOW?)

Step 5 : Create selection criteria to compare / assess these concepts

Step 6 : Use the selection criteria to assess/rank these concepts

Scale : 1-5 (5 = best or most attractive)

Selection criteria	Concept 1: Daily de-humidifier	Concept 2 Floating Iceberg	Concept 3 Flying Iceberg
Feasibility of concept	4	3	2
Cost of implementation	4 (lowest cost)	2	1
Quality (of water)	5	5	5
Reliability	3	2	1
...			

Using the selection criteria, and the assessment in the table above, the "best" concept is the "Daily de-humidifier", which was used as the basis for the start-up.

2. Product Strategy (PS)

PS contains the following components:

1. The new product concept (see above)
2. The resources needed to develop the product
"what?" {
 - Expertise: people, skill-sets
 - Infrastructure → space, prototyping facilities, computing equipment,

- Source ("where?") → from where are we going to obtain the necessary resources?
"In-house?", "Out-source?",
"Hybrid?"

- Timing ("when?") → when are we going to acquire the necessary resources
⇒ Create a plan or schedule for obtaining the necessary resources.

2. Vision, Mission, and Business Goals

Examples

① Microsoft (around 1986, when MS was a start-up)

Vision (1980s) : - a computer in every home, and at every desk (in a company)

Mission : Every one of these computers will use a MS Operating System (OS) to manage all computing tasks (in a computer)

② Google (around 2000/2001 when Google was a start-up)

Vision (~2000) : all the information in the world available to everyone at any time.

Mission : To organize the world's information & create software tools (like search engines) to make information available.....

Definitions:

Vision: a high-level view of what the future might look like,
e.g., a society with a robot in every home.

Mission: how a particular company will contribute to that vision

e.g., Company X's mission might be to create sensors (touch, feel, see) to help make that vision a reality.